Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)			**Indicative content** Elements and water • reactivity increases down **both** groups • both potassium and calcium metals react with water to give the hydroxide and hydrogen • Group 1 metals react more vigorously • Group 1 metals lose only one electron while Group 2 metals lose two electrons / Group 1 metals form cations easier than Group 2 metals • potassium melts into a ball and catches fire/lilac flame • calcium produces a steady stream of bubbles/cloudy white solution forms • $2K + 2H_2O \rightarrow 2KOH + H_2$ • $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$ Carbonates • Group 1 and Group 2 carbonates show different properties • Group 1 carbonates are soluble, Group 2 carbonates are insoluble **5-6 marks** Comparison of similarities/differences, description of reactions and appropriate equations *The candidate constructs a relevant, coherent and logically structured method including all key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.* **3-4 marks** Comparison of similarities/differences and description of reactions OR description of reactions and appropriate equations *The candidate constructs a coherent account including most of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary are generally sound.*	4	2		6		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				1-2 marks Description of reactions only or fair attempt at equations *The candidate attempts to link at least two relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.* **0 marks** *The candidate does not make any attempt or give an answer worthy of credit.*						
	(b)			$n(CO_2) = \frac{186}{24500} = 7.59 \times 10^{-3}$ (1) $n(K_2CO_3) = 7.59 \times 10^{-3}$ mass $K_2CO_3 = 7.59 \times 10^{-3} \times 138.2 = 1.05$ g (1) mass impurity = 1.40 – 1.05 = 0.35 g (1)		2	1	3	2	
	(c)			flame test - lilac flame (1) (HNO_3 followed by) $AgNO_3(aq)$ - white precipitate (1)	2			2		2
	(d)			large increase from 1st to 2nd ionisation energy / high 2nd ionisation energy / too much energy needed to remove 2nd electron (1) second electron removed from a shell nearer to the nucleus / with increased effective nuclear attraction / less shielding (1)	2			2		
				Question 10 total	**8**	**4**	**1**	**13**	**2**	**3**